

The Effects of Age, Gender, and Exercise Type on Physiological Stress Markers in Islander Populations

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Stats 101B Lecture 3 - Group L

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Abstract

Understanding how the human body responds to physical exertion is essential for designing effective fitness regimens, managing stress, and tailoring health interventions. In this study, we investigate how exercise type, age group, and gender influence physiological stress markers (namely adrenaline, cortisol, and blood oxygen levels) in a representative sample of the Islander population. We hypothesize that high-intensity activities would elicit stronger physiological responses, and that these responses would vary based on demographic factors. Participants engaged in one of three activities (bungee jumping, swimming, or light jogging), and changes in hormone and oxygen levels were measured before and after exercise. The experimental design was analyzed using a series of three-way factorial ANOVA to evaluate main effects and interaction effects.

Introduction

Physical activity exerts widespread influence on the human body, playing a critical role in regulating stress, hormonal balance, and long-term health outcomes. One of the body's primary hormonal responses to physical and psychological stress is the release of cortisol, a glucocorticoid produced by the adrenal cortex. While often misunderstood as harmful due to its catabolic effects, cortisol actually also serves essential roles in energy metabolism, immune function, and adaptation to stressors such as exercise (Hackney & Walz, 2013). Additionally, adrenaline, which is secreted rapidly in response to acute stress, enhances cardiovascular output and energy availability. This highlights how these hormones operate as coordinated components of the body's stress response system.

Research has also shown that the intensity of physical activity directly influences cortisol release, with moderate to high intensity exercise (defined as $\geq 60\%$ VO₂ max) triggering substantial increases, while low intensity activity may result in little (or even negative) changes in cortisol levels (Hill et al., 2008). Moreover, regular physical activity has been linked to improved sleep quality and reductions in circulating cortisol, underscoring its dual benefit for both stress regulation and recovery (De Nys et al., 2022). However, these existing studies focus on the clinical population with limited data on healthy older adults or males, which leaves gaps in our understanding of how cortisol and physical activity interact across diverse demographic groups.

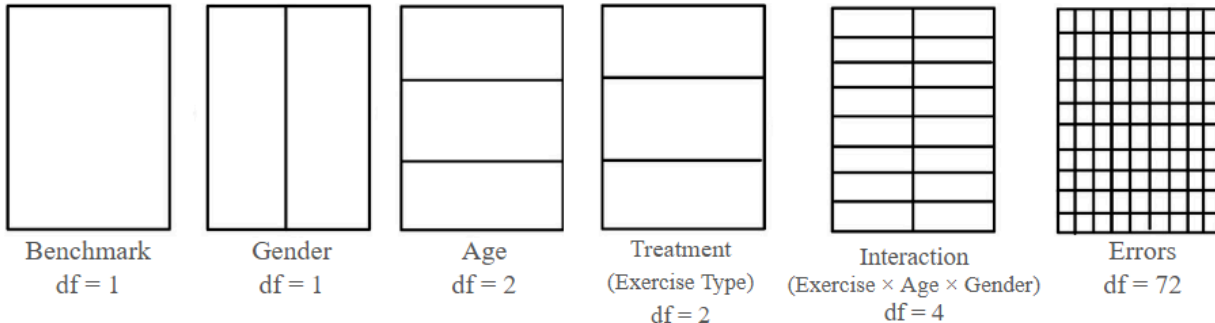
Age and gender are two crucial yet overlooked modifiers of exercise response. As people age, their physical activity levels decline significantly by as much as 40% to 80%, due to loss of muscle mass, chronic illness, and cognitive/social factors, which may also in turn affect hormonal responses and endurance capacity (Suryadinata et al., 2020). Additionally, while men typically possess higher VO₂ max values due to anatomical and physiological advantages, women have been shown to pace more evenly, utilize fat more efficiently, and experience less fatigue during endurance events (Besson et al., 2022).

Despite these known influences, few studies have attempted to identify clear patterns in how physical activity type, age, and gender interact with cortisol, adrenaline, and VO2 (endurance) responses. Understanding these patterns is important not only for optimizing individual training and recovery strategies but also for informing public health interventions, especially for senior or at-risk populations. In this study, we aim to address this gap by analyzing hormone responses to different physical activities across age and gender groups within Islander populations, to explore whether measurable relationships exist between demographic factors, activity type, and hormonal stress responses.

Methods

Participants

This study analyzed a dataset collected from a representative sample of individuals from the Islander population. A total of 90 participants were included, comprising an equal split of both male and female individuals across three age groups: 0–25 years, 26–50 years, and 51+ years. All participants were generally healthy and engaged in one of three physical activities as part of the study protocol. Demographic data, including age, gender, and island of origin, were recorded for all participants.



Experimental Design

This study employed a between-subjects, fully-crossed factorial design to examine the effects of three independent variables—exercise type (bungee jumping, light jogging, swimming), age group (0–25, 26–50, 51+), and gender (male, female)—on three dependent physiological stress markers: adrenaline, cortisol, and blood oxygen levels. Each participant was randomly assigned to one of the three exercise conditions and completed only one activity to ensure independence of observations and avoid carryover effects between exercise types.

A total of 90 participants were recruited and evenly distributed across all 18 experimental conditions (3 age groups × 2 genders × 3 exercise types), with five individuals per unique combination. This structure enabled the use of three-way ANOVA to assess both main effects

and interaction effects across factors. The exercise conditions were selected to represent a range of intensities: bungee jumping (high intensity), swimming (moderate intensity), and light jogging (low intensity).

Each participant had their adrenaline, cortisol, and oxygen levels measured immediately before and after exercise, and the difference between pre- and post-activity levels was computed to derive change scores for statistical analysis. All measurements were conducted using standardized clinical instruments, ensuring consistency in hormone quantification and blood oxygenation readings.

The study was conducted using an open-label design, where both participants and researchers were aware of the assigned exercise condition. This design was considered acceptable due to the objective nature of the outcome measures and the physical impossibility of blinding participants to their assigned exercise type. All procedures and data collection protocols were standardized across participants to reduce variability unrelated to the experimental manipulations.

Instruments

Physiological measurements were taken immediately before and after exercise using standard clinical tools:

- Blood samples were used to determine adrenaline and cortisol levels via immunoassay.
- Oxygen levels were assessed using blood gas analyzers that measured partial pressure of oxygen (mmHg).
- All hormone values were recorded in standardized units, and delta (change) scores were calculated by subtracting pre-exercise values from post-exercise values.
- Additional variables (e.g., name, island, and exercise type) were recorded for descriptive and grouping purposes.

Procedure

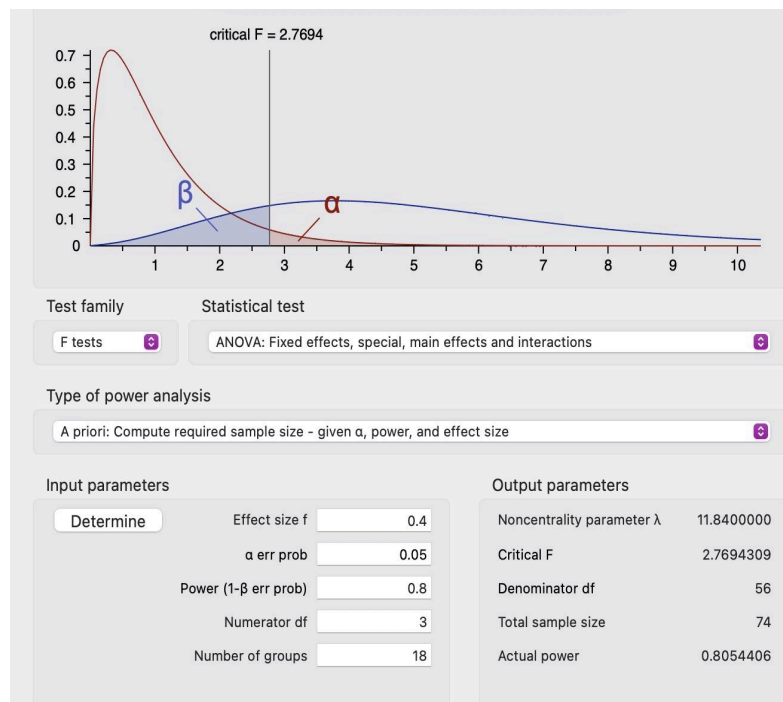
Participants were assigned to one of the three physical activity conditions: bungee jumping (50 m), light jogging (5 minutes), or swimming (50 m). Prior to the exercise, a baseline blood sample was collected to measure cortisol, adrenaline, and oxygen levels. After completing the activity, a second blood sample was drawn to assess post-exercise hormone levels.

Each participant's age group and gender were noted, and the data were anonymized prior to analysis. Hormone changes were computed for each individual, and the data were cleaned to remove missing or non-physiological values. Statistical analysis was conducted using three-way ANOVA to assess the main effects and interactions of exercise type, age group, and gender on each physiological outcome.

Data Analysis

Sample Size Determination

To determine the required sample size, we used GPower and selected a power level of 0.80, meaning there is an 80% probability of correctly rejecting a false null hypothesis. We used an alpha level of 0.05, the standard threshold for statistical significance, which represents the probability of making a Type I error (i.e., falsely rejecting a true null hypothesis). Based on preliminary analysis, we estimated a conservative effect size of 0.40, which, due to its relatively large magnitude, led to a reduced sample size requirement, favorable for the feasibility of the study. Given our three-way factorial design, the largest degrees of freedom across main effects and interactions is 3.



According to G*Power calculations for this design, the minimum total sample size required is 74. However, to ensure a balanced and fully crossed design, we adjusted upward to accommodate 5 observations per condition (i.e., per unique combination of gender $\times$ age group $\times$ exercise type). With 2 genders $\times$ 3 age groups $\times$ 3 activities = 18 unique groups, a minimum of 5 participants per group results in a total sample size of 90.

This allocation yields 45 participants per gender, with 15 individuals per age bracket (0–25, 26–50, 51+). Each unique subgroup (e.g., females aged 0–25 doing bungee jumping) includes 5 participants, ensuring independence of groups and allowing the study to conform to the requirements of the factorial design. Notably, we did not assign participants to perform all three

activities, so as to preserve the independence of each observation and match the assumptions of our between-subjects model.

Statistical Analysis

Using R, we performed a series of three-way between-subjects ANOVAs to examine the effects of exercise type, age group, and gender on each physiological stress marker: adrenaline change, cortisol change, and oxygen change. This factorial design allowed us to test for significant main effects of each independent variable and explore interaction effects between them. The study specifically investigated whether different forms of physical activity, bungee jumping, swimming, or light jogging, led to varying hormonal responses, and whether these responses were moderated by age group (0–25, 26–50, 51+) and gender (male, female).

Each participant completed each exercise, with recovery time in between, to preserve independence, with 5 participants per group across 18 unique combinations, totaling 90 individuals evenly split across gender and age brackets. Blood samples were taken before and after activity to calculate changes in physiological measures. All participants and researchers were aware of assigned activities (open-label design).

F-tests were used to assess the statistical significance of main and interaction effects, and Tukey's Honest Significant Difference (HSD) tests were conducted as post-hoc analyses to probe significant group differences. Analyses were conducted at a significance level of $\alpha = 0.05$.

This approach enabled us to evaluate not only whether exercise type affects biological stress markers, but also how these effects vary across demographic subgroups. These insights are meant to contribute to our understanding of how stress responses differ across age and gender within the Islander population, especially in the context of high-exertion activity like swimming.

Results

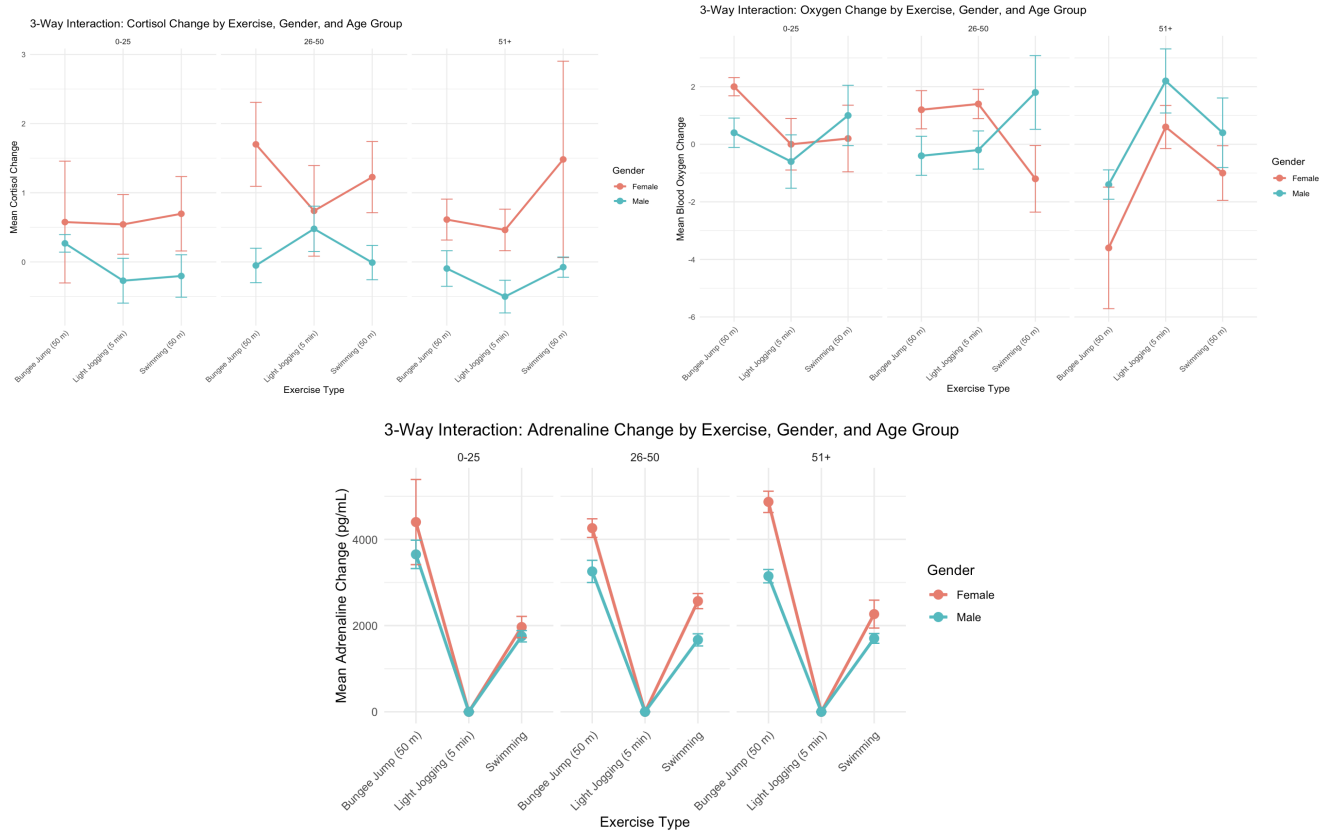


Figure 1. Three-way interaction plots of mean changes in oxygen, cortisol, and adrenaline levels by exercise type, gender, and age group.

Figure 1 presents three-way interaction plots for the changes in oxygen, cortisol, and adrenaline levels by exercise type, gender, and age group. Across all three biomarkers, light jogging induced minimal changes that hovered near zero, regardless of demographic subgroup. In contrast, swimming and bungee jumping resulted in more substantial physiological changes. Notably, older females (51+) exhibited the largest increases in adrenaline, especially following bungee jumping. For our study to be realistic and widely applicable, we decided to focus our attention on only swimming as the form of exercise, as it is the most probable form of exercise for the average person to engage in.

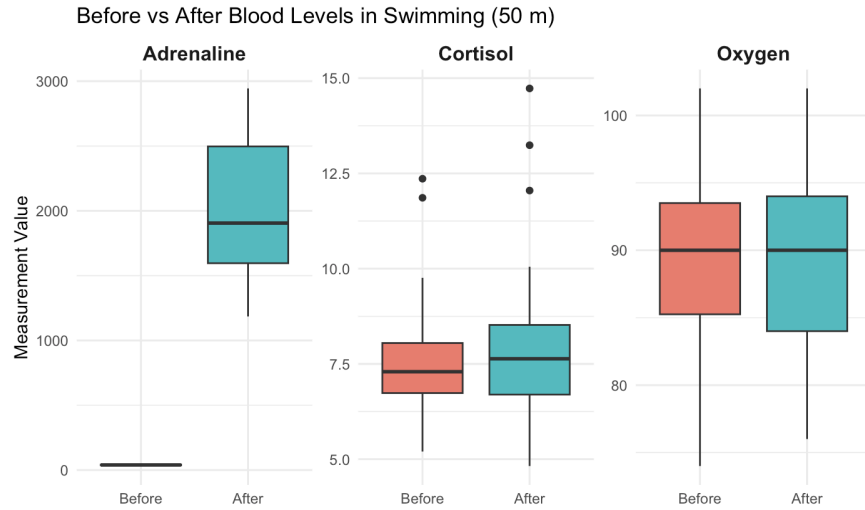


Figure 2. Pre- and post-exercise boxplots for adrenaline, cortisol, and oxygen levels during swimming (50 m).

Figure 2 shows pre- and post-swimming boxplots for each biomarker. Adrenaline levels rose sharply after swimming, with males experiencing the most pronounced increases. Cortisol levels also rose post-swim but with higher variability and a more modest effect size. Oxygen levels remained stable overall, showing minimal changes between pre- and post-conditions.

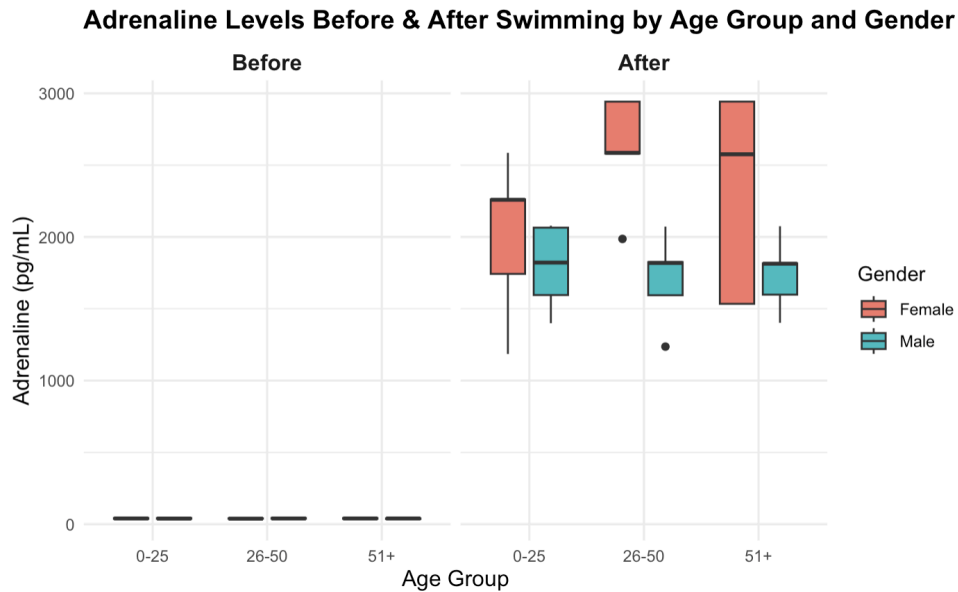


Figure 3. Adrenaline levels before and after swimming by age group and gender.

Pre-exercise levels were near zero across all groups. Post-swimming, females—especially those aged 51+—had the highest adrenaline responses, with males consistently showing lower and less variable increases.

Figure 3 breaks down adrenaline levels before and after swimming by age group and gender. Pre-exercise levels were consistently near zero. Post-swimming, females in the 51+ group showed the most substantial adrenaline increases, while males across all age groups exhibited smaller and less variable changes.

ANOVA Table: Cortisol Change by Age Group and Gender						ANOVA Table: Adrenaline Change by Age Group and Gender					
Term	DF	SumSq	MeanSq	Statistic	P-Value	Term	DF	SumSq	MeanSq	Statistic	P-Value
Age Group	2	1.163	0.582	0.254	0.776	Age Group	2	334176.9	167088.5	0.813	0.455
Gender	1	11.371	11.371	4.962	0.036	Gender	1	2345117.0	2345117.0	11.412	0.002
Age Group and Gender	2	0.541	0.271	0.118	0.889	Age Group and Gender	2	587669.4	293834.7	1.430	0.259
Residuals	24	55.001	2.292	NA	NA	Residuals	24	4932083.5	205503.5	NA	NA

ANOVA Table: Blood Oxygen Change by Age Group and Gender					
Term	DF	SumSq	MeanSq	Statistic	P-Value
Age Group	2	4.200	2.100	0.324	0.726
Gender	1	22.533	22.533	3.476	0.075
Age Group and Gender	2	6.467	3.233	0.499	0.613
Residuals	24	155.600	6.483	NA	NA

Figure 4. ANOVA tables for cortisol, adrenaline, and oxygen changes by age group and gender.

Gender was significant for both cortisol ($p = 0.036$) and adrenaline ($p = 0.002$), but not for oxygen. Age and interaction effects were nonsignificant across all models.

ANOVA tables confirmed that:

- Gender was a significant factor for:
 - Cortisol ($p = 0.036$)
 - Adrenaline ($p = 0.002$)
- Age group and age $\times$ gender interactions were not significant for any biomarker (all $p > 0.27$).
- Oxygen changes were not significantly affected by any factor.

MANOVA Table: Blood Change Overall by Age Group and Gender						
Term	DF	Pillai	Statistic	Num DF	Den DF	P-Value
Age Group	2	0.125	0.512	6	46	0.796
Gender	1	0.484	6.868	3	22	0.002
Age Group and Gender	2	0.161	0.672	6	46	0.673
Residuals	24	NA	NA	NA	NA	NA

Figure 5. MANOVA table summarizing the multivariate analysis of blood marker changes.

Only gender had a statistically significant multivariate effect ($p = 0.002$), confirming its overall influence. Age and interaction terms were not significant.

A follow-up MANOVA supported these results, revealing a significant multivariate effect of gender ($p = 0.002$) across all three biomarkers. Neither age nor interaction effects were significant.

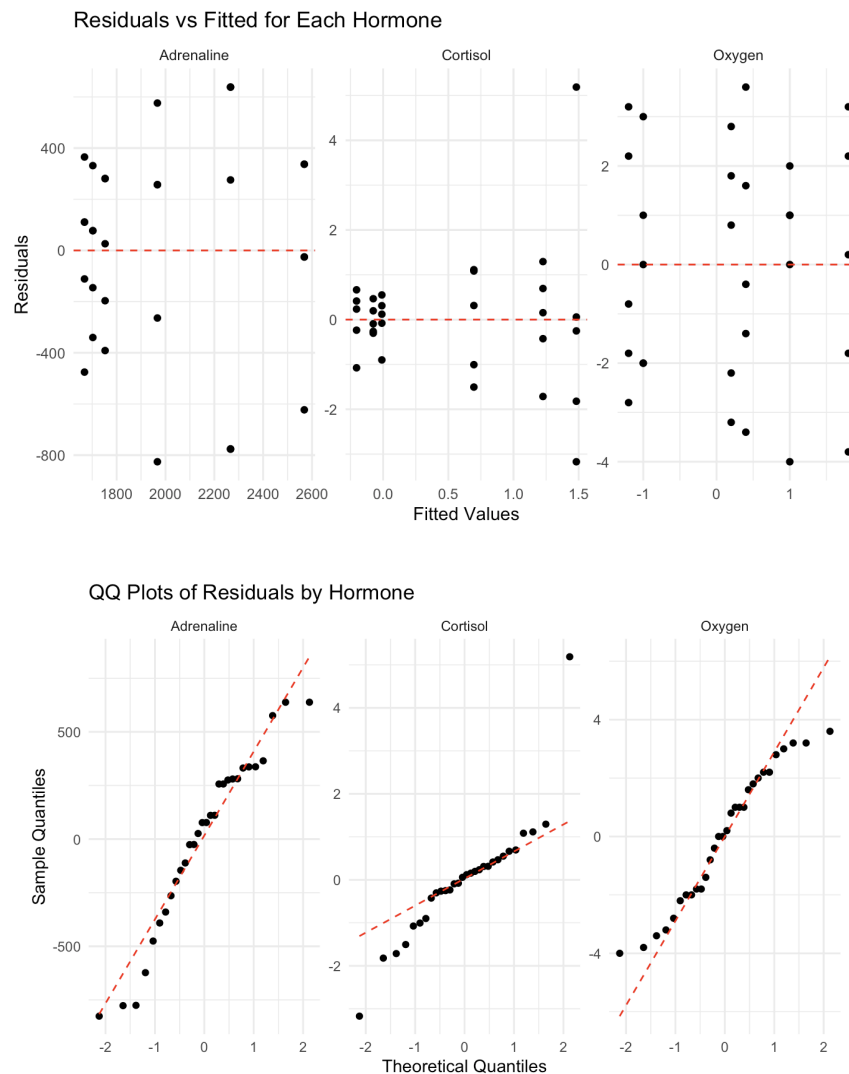


Figure 6. *Residual vs. fitted and Q-Q plots for adrenaline, cortisol, and oxygen models.*

Residual plots showed no systematic patterns, indicating homoscedasticity. Q-Q plots for all three biomarkers demonstrated approximate normality, supporting the validity of model assumptions.

Figure 6 displays Q-Q plots and residual vs. fitted plots for all biomarkers. Residuals showed no concerning patterns, suggesting homoscedasticity, and Q-Q plots indicated approximate normality, supporting the assumptions of ANOVA.

Tukey HSD Test: Cortisol Change by Age Group					Tukey HSD Test: Blood Oxygen Change by Age Group				
	Diff	Lower	Upper	P-Adj		Diff	Lower	Upper	P-Adj
25-50 and 0-25	0.362	-1.329	2.053	0.855	25-50 and 0-25	-0.3	-3.144	2.544	0.963
51+ and 0-25	0.457	-1.234	2.148	0.780	51+ and 0-25	-0.9	-3.744	1.944	0.712
51+ and 26-50	0.095	-1.596	1.786	0.989	51+ and 26-50	-0.6	-3.444	2.244	0.859

Tukey HSD Test: Adrenaline Change by Age Group				
	Diff	Lower	Upper	P-Adj
25-50 and 0-25	258.47	-247.813	764.753	0.423
51+ and 0-25	124.60	-381.683	630.883	0.814
51+ and 26-50	-133.87	-640.153	372.413	0.788

Figure 7. Tukey HSD test results by age group for cortisol, oxygen, and adrenaline change.

No statistically significant differences were detected between age groups for any of the three biomarkers ($p_{adj} > 0.7$ for all comparisons).

Tukey HSD Test: Cortisol Change by Gender Group					Tukey HSD Test: Blood Oxygen Change by Gender Group				
	Diff	Lower	Upper	P-Adj		Diff	Lower	Upper	P-Adj
Male-Female	-1.231	-2.372	-0.09	0.036	Male-Female	1.733	-0.186	3.652	0.075

Tukey HSD Test: Adrenaline Change by Gender Group				
	Diff	Lower	Upper	P-Adj
Male-Female	-559.18	-900.819	-217.541	0.002

Figure 8. Tukey HSD test results by gender for cortisol, oxygen, and adrenaline change.

Figure 7 and Figure 8 present Tukey HSD results by age and gender:

- By age group, no statistically significant differences were found for cortisol, oxygen, or adrenaline (all adjusted p -values > 0.70).
- By gender, significant differences were observed:
 - Cortisol: males had lower increases than females ($p = 0.036$).
 - Adrenaline: males had significantly smaller changes than females ($p = 0.002$).
 - Oxygen: marginal difference by gender ($p = 0.075$), with males trending higher.

Discussion

The study initially explored how physiological stress responses, measured via changes in adrenaline, cortisol, and oxygen, varied across different exercise types, age groups, and genders. Preliminary results showed that light jogging produced negligible hormonal changes across all demographics, prompting a focus on the more dynamic responses induced by swimming and

bungee jumping. Given the broader accessibility and real-world relevance of swimming, the final analysis concentrated on pre- and post-swimming changes, with an emphasis on two-way interactions between age group and gender.

Among the three biomarkers, adrenaline demonstrated the most pronounced response to swimming, with substantial increases observed post-exercise. This effect was especially strong in females aged 51+, who exhibited the highest post-swim adrenaline levels (Figure 3). Cortisol also showed a modest increase following swimming, though with greater variability, while oxygen levels remained relatively stable, showing no significant change between pre- and post-exercise measurements (Figure 2).

Statistical testing corroborated these observations. Two-way ANOVAs indicated that gender significantly influenced both cortisol ($p = 0.036$) and adrenaline responses ($p = 0.002$), with females exhibiting greater increases than males. Tukey HSD tests confirmed these gender-based differences and showed no significant effects by age group for any biomarker ($p_{\text{adjusted}} > 0.7$ across all comparisons) (Figures 7–8). Although oxygen changes approached marginal significance by gender ($p = 0.075$), this effect was not statistically conclusive. A MANOVA further validated these findings, identifying gender as the only significant multivariate predictor of combined biomarker changes ($p = 0.002$), while age and interaction terms remained nonsignificant (Figure 5).

Model diagnostics supported the robustness of these conclusions. Q-Q plots for each response variable revealed approximate normality, while residual vs. fitted plots showed no systematic patterns, indicating homoscedasticity and appropriate model fit (Figure 6).

In summary, the findings highlight gender as a key determinant in physiological stress response to swimming, with females, particularly in the oldest age group, demonstrating elevated hormonal reactivity. These results underscore the importance of considering demographic factors when evaluating biological responses to physical activity and may inform future research or fitness programming tailored to population subgroups.

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